

How Much Can We Compress Small LLMs? Quantization Trade-offs for Low-Resource Bangla Language Understanding

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Abstract

Large language models (LLMs) offer strong multilingual capabilities, but their memory requirements hinder deployment in resource-constrained settings, particularly for low-resource languages. Although post-training quantization can reduce these requirements, its effects on low-resource language understanding remain insufficiently characterized. We present a controlled study of model capacity and quantization for Bangla language understanding using Qwen2.5 models at 0.5B, 1.5B, and 3B parameters under FP16, INT8, and 4-bit NormalFloat (NF4) across stance classification, natural language inference, and fake-news classification, with Falcon3-3B providing cross-family validation. Qwen2.5-3B FP16 achieves Macro-F1 scores of 0.435, 0.728, and 0.664 across the three tasks, while the 0.5B model exhibits class collapse even before quantization. INT8 retains 96.2–99.9% of Qwen2.5-3B performance, whereas NF4 retention ranges from 77.3–91.3%. On an NVIDIA Tesla T4, INT8 and NF4 reduce the Qwen2.5-3B footprint by 45.0% and 67.4%, respectively, but neither improves inference speed. These findings show that effective LLM compression for low-resource languages requires jointly considering model capacity, task sensitivity, memory efficiency, and deployment hardware. Code is available at [GitHub](#).

1 Introduction

Large language models (LLMs) have demonstrated strong capabilities across a wide range of natural language processing tasks, including text classification (Pecher et al., 2026), question answering (Singh et al., 2025), reasoning (Li et al., 2026), and instruction following (Jia et al., 2026). Recent open-weight model families have further expanded access to instruction-tuned LLMs at different parameter scales and across multiple languages (Chen et al., 2024; Wang et al., 2024). Qwen2.5, for example, provides models ranging

from compact to large-scale variants with multilingual and instruction-following capabilities (Qwen et al., 2025). Despite these advances, the computational and memory requirements of LLMs remain a major barrier to their use in resource-constrained settings (Kim et al., 2024; Matarazzo and Torlone, 2025).

This challenge is particularly relevant to multilingual and low-resource language processing (Khade et al., 2025; Joshi et al., 2025). Languages such as Bangla remain comparatively underrepresented in widely used multilingual resources and evaluation settings, while access to high-end computational infrastructure can also be limited (Hosain and Morol, 2025; Hasan et al., 2024). Smaller multilingual LLMs provide a more practical alternative, but their capacity is already constrained, and further compression may affect low-resource language understanding differently across model scales and tasks (Ahmed et al., 2025). Consequently, compression strategies developed and evaluated primarily in broader or high-resource settings cannot automatically be assumed to preserve Bangla language-understanding capabilities (Haider et al., 2025).

Post-training quantization (PTQ) provides a practical approach to reducing the computational footprint of LLMs by representing model weights at lower numerical precision (Zhao et al., 2025). INT8 quantization can substantially reduce memory requirements while preserving much of the original model capability (Dettmers et al., 2022), while 4-bit NormalFloat (NF4) enables more aggressive compression (Dettmers et al., 2023). However, the effect of quantization can depend on model capacity, task characteristics, model family, and hardware (Li et al., 2024). In particular, reducing model footprint does not necessarily guarantee preserved downstream performance or faster inference (Huang et al., 2024). **This raises our central research question:** *How much can small multilingual LLMs be compressed before*

losing substantial performance on low-resource Bangla language-understanding tasks, and what performance-efficiency trade-offs emerge across model scales and tasks?

To answer this question, we conduct a controlled study of Qwen2.5-Instruct models at 0.5B, 1.5B, and 3B parameters under FP16, INT8, and 4-bit NF4 precision across three Bangla language-understanding tasks: stance classification, natural language inference (NLI), and fake-news classification. We further evaluate Falcon3-3B-Instruct under the same conditions as a matched cross-family comparison, resulting in 36 model-precision-task evaluations across four models, three precision settings, and three tasks. In addition, we benchmark model footprint, inference memory, latency, and throughput on an NVIDIA Tesla T4. By jointly examining model capacity, quantization, task sensitivity, and computational efficiency, our study distinguishes baseline capacity limitations from compression-induced degradation and provides a multidimensional assessment of LLM compression for low-resource Bangla language understanding. Ultimately, our goal is to identify when compression offers a practical balance between language-understanding performance and limited computational resources.

2 Related Work

2.1 LLM Quantization and Efficiency

Quantization has become a widely used approach for reducing the memory and computational requirements of LLMs. Early methods demonstrated that lower-precision inference can retain strong model performance: LLM.int8() introduced mixed-precision decomposition for 8-bit inference (Dettmers et al., 2022), while SmoothQuant enabled training-free W8A8 quantization by mitigating activation outliers (Xiao et al., 2023). QLoRA subsequently introduced 4-bit NormalFloat (NF4), demonstrating the effectiveness of 4-bit representations for memory-efficient LLM adaptation (Dettmers et al., 2023).

Recent work has shifted toward understanding when such compression remains effective. Large-scale evaluations show that quantization behavior varies across model families, numerical formats, calibration settings, and downstream tasks (Li et al., 2024; Gong et al., 2024; Kurtic et al., 2025). This variation becomes particularly important for small models: Sk et al. (2025) studied performance–

efficiency trade-offs in models as small as 0.5B parameters, while Wang et al. (2025) showed that SLMs can exhibit compression behavior distinct from larger LLMs. Quantization sensitivity also increases in challenging settings; long-context evaluation reveals substantially greater degradation for some 4-bit configurations than for 8-bit quantization (Mekala et al., 2025). Similarly, recent evaluation of microscaling formats reports near-lossless MXFP8 performance but greater degradation under MXFP4 (Zhang et al., 2026). Together, these findings suggest that compression must be evaluated jointly with model scale, task characteristics, and efficiency rather than through bit width alone.

2.2 Multilingual and Low-Resource Compression

Most quantization research has historically centered on English, although compression effects do not transfer uniformly across languages. Marchisio et al. (2024) showed that languages can be disproportionately affected by quantization, with non-Latin-script languages exhibiting particularly strong degradation in some settings. This has motivated language-aware compression strategies. Zeng et al. (2024) demonstrated that multilingual calibration can better preserve low-resource language performance, while subsequent work showed that non-English and multilingual calibration data can outperform English-only calibration for multilingual LLMs (Chimoto et al., 2026). Related work on multilingual encoder compression further found that performance retention depends on the amount of language-specific information available to the model (Gurgurov et al., 2025). These findings indicate that English-centric compression results cannot be assumed to generalize directly to low-resource languages.

2.3 LLMs for Bangla Language Understanding

Bangla remains comparatively underexplored in LLM evaluation. BenLLM-Eval showed that although zero-shot LLMs can perform competitively on some Bangla tasks, performance remains weak on others (Kabir et al., 2024). More recent studies similarly identify challenges in Bengali processing, including inefficient tokenization and inconsistent performance across downstream tasks (Mahfuz et al., 2025). TituLLMs addressed this direction through Bangla-oriented 1B and 3B models and dedicated benchmarking resources, while

also demonstrating that language adaptation does not uniformly improve performance (Nahin et al., 2025). Hossain et al. (2026) evaluated quantized 7B–20B models across five Bangla NLU benchmarks and found substantial model- and quantizer-dependent variation. However, compression behavior at substantially smaller model scales remains less understood for Bangla.

3 Experimental Setup

3.1 Datasets and Tasks

In this study, we evaluate the models on three Bangla language-understanding datasets: the *Bangla July Uprising Stance Dataset (BJUS)* for stance classification, *Bengali Cross-Lingual Natural Language Inference Dataset (XNLI_BN)* for natural language inference, and *BanFakeNews* for fake-news detection. We evaluate all model and quantization configurations on the same fixed test partitions to enable consistent, controlled comparisons across experimental settings (See Table 1).

BJUS is an extended version of the Student-People Mass Uprising Public Sentiments Dataset (Hossen et al., 2025), incorporating additional Bangla comments and new stance-based labels. The dataset contains comments related to the July Revolution in Bangladesh, categorized into three stance classes: **Pro-Uprising**, **Neutral**, and **Anti-Uprising**. Comments expressing support for the uprising are labeled as Pro-Uprising, those opposing the uprising as Anti-Uprising, and those expressing neither position as Neutral. The task is formulated as stance classification rather than conventional sentiment classification, as emotional polarity does not necessarily reflect a commenter’s position toward the uprising. Further details on the dataset extension and labeling procedure are provided in Appendix A.1.

XNLI-BN is a Bangla natural language inference dataset (Bhattacharjee et al., 2021), curated from the subset of MultiNLI (Williams et al., 2018) used in XNLI (Conneau et al., 2018) and machine-translated from English into Bangla. Translation quality was filtered using LaBSE similarity, with low-similarity instances removed. Each premise–hypothesis pair is categorized as **Entailment**, **Neutral**, or **Contradiction**. Since multiple hypotheses may share the same premise, we use group-aware splitting to keep all examples associated with a premise within the same partition, prevent-

ing premise-level overlap across the train, development, and test sets. Detailed dataset preprocessing is provided in Appendix A.2.

BanFakeNews is an annotated Bangla fake-news dataset (Hossain et al., 2020). The articles are categorized as **Authentic** or **Fake** and include information such as headlines, article content, and publication metadata. We use the headline and article content as input for binary fake-news classification. We retain the original binary label mapping, with 1 denoting Authentic and 0 denoting Fake.

Dataset	Train	Dev	Test	Total
BJUS	7,853	982	982	9,817
XNLI-BN	10,024	1,251	1,253	12,528
BanFakeNews	6,800	850	851	8,501

Table 1: Experimental Dataset Statistics.

3.2 Models

We use Qwen2.5-Instruct (Qwen et al., 2025) as the primary model family, evaluating three parameter scales: 0.5B, 1.5B, and 3B. Models from the same family are intentionally selected to provide a controlled comparison across model scales while minimizing variation arising from differences in model architecture, tokenizer, and training procedure. This allows us to examine the interaction between model capacity and quantization more directly. To assess whether the observed compression behavior is specific to Qwen2.5, we additionally evaluate Falcon3-3B-Instruct (Falcon-LLM Team, 2024) as a cross-family comparison at the 3B scale. Table 2 summarizes the models used in our experiments.

Model	Parameters	Model Identifier
Qwen2.5-Instruct	0.5B	Qwen/Qwen2.5-0.5B-Instruct
Qwen2.5-Instruct	1.5B	Qwen/Qwen2.5-1.5B-Instruct
Qwen2.5-Instruct	3B	Qwen/Qwen2.5-3B-Instruct
Falcon3-Instruct	3B	tiuae/Falcon3-3B-Instruct

Table 2: Models evaluated in this study.

The selected Qwen2.5 scales provide a progression from a highly compact model to a more capable small language model while remaining suitable for resource-constrained inference. The Falcon3-3B comparison further allows us to examine whether compression effects observed at the 3B scale generalize across model families

3.3 Quantization Configurations

Each model is evaluated under three numerical precision configurations: FP16, INT8 (Dettmers et al.,

2022), and 4-bit NF4 (Dettmers et al., 2023). FP16 serves as the unquantized reference, while INT8 and 4-bit NF4 represent progressively more aggressive compression settings. The same quantization configurations are applied across models to enable controlled comparison of performance and efficiency under reduced precision. Table 3 summarizes the configurations used in our experiments.

Configuration	Weight Precision	Implementation
FP16	16-bit	Half precision
INT8	8-bit	bitsandbytes
4-bit NF4	4-bit	bitsandbytes NF4

Table 3: Experimental Quantization Configurations

FP16 provides the reference performance for each model. INT8 quantization is implemented using the bitsandbytes integration in Hugging Face Transformers. For the most aggressive compression setting, we use 4-bit NormalFloat (NF4) quantization with FP16 computation and without double quantization. Although this configuration is internally identified as int4 in our implementation, it corresponds to NF4 weight quantization rather than integer INT4 arithmetic.

3.4 Inference Setup

Models and tokenizers were loaded using Hugging Face Transformers, with quantized configurations implemented through bitsandbytes. All official experiments were conducted on an NVIDIA Tesla T4 GPU in the Kaggle environment using CUDA-enabled inference. Models were evaluated in inference mode without parameter updates. Thus, our experiments study post-training quantization (PTQ) rather than quantization-aware training or parameter-efficient fine-tuning.

3.5 Prompting and Inference Protocol

We formulate all tasks as constrained generation, using task-specific instructions and fixed label sets. Prompts are kept identical across all model and quantization configurations. Inference is deterministic with sampling disabled and max_new_tokens=8. The maximum input length is fixed at 4,096 tokens; longer inputs are truncated while preserving the task instruction and prompt structure. Generated responses are mapped to class labels using task-specific parsers, while unparsable outputs are recorded as parsing failures. Further details are provided in Appendix B.

3.6 Evaluation and Performance Analysis

Each model–quantization configuration is evaluated independently on the complete fixed test partition of each task. The same evaluation pipeline, prompts, generation settings, and parsing procedure are used across all configurations to ensure controlled comparison. No subsampling is performed during the official evaluations. Development and smoke-test outputs are stored separately from the final results to prevent partial predictions from being mixed with complete evaluation outputs.

3.6.1 Evaluation Metrics

Macro F1 is used as the primary performance metric. For a classification problem with C classes, Macro F1 is defined as

$$F1_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C F1_c, \quad (1)$$

where $F1_c$ denotes the F1 score for class c . Macro F1 assigns equal importance to each class regardless of its frequency, making it particularly suitable for our setting, where class distributions are uneven and prediction collapse toward a dominant class may not be adequately reflected by accuracy alone.

For each configuration, we additionally report accuracy, macro precision, macro recall, per-class F1, and parse success rate. Accuracy is treated as a secondary performance measure, while per-class F1 and parse success are retained for diagnostic analysis.

3.6.2 Performance Retention Analysis

To quantify the effect of compression, each quantized configuration was compared against the FP16 baseline of the same model and task.

Performance retention was calculated as:

$$\text{Retention} = \frac{F1_{\text{quantized}}}{F1_{\text{FP16}}} \times 100 \quad (2)$$

where $F1_{\text{quantized}}$ represents the Macro F1 obtained under INT8 or 4 bit NF4 quantization.

Absolute performance change was also calculated as:

$$\Delta F1 = F1_{\text{quantized}} - F1_{\text{FP16}} \quad (3)$$

This analysis makes it possible to distinguish absolute task performance from robustness to compression. A model may achieve a lower overall

F1 while still retaining most of its baseline performance after quantization, whereas another model may have stronger baseline performance but experience a larger relative degradation.

3.6.3 Efficiency Benchmarking

A separate controlled benchmark was conducted to measure the computational behavior of FP16, INT8, and 4-bit NF4 configurations. All configurations were benchmarked using the same GPU environment and standardized inference conditions. A fixed input length of approximately 512 tokens was used with a batch size of 1 and a maximum generation length of 8 tokens.

Before collecting timing measurements, three warm-up runs were performed to reduce the influence of initialization effects. Each configuration was subsequently measured over 20 inference runs. CUDA synchronization was performed around each timed inference region to ensure that asynchronous GPU execution did not produce artificially low latency measurements.

The benchmark recorded model loading time, allocated GPU memory, model footprint, peak inference memory, mean and median inference latency, examples processed per second, and generated tokens per second. Model footprint was used as the primary indicator of memory reduction because allocator-based GPU memory measurements can vary with runtime behavior. Latency and throughput measurements were used to characterize the inference-efficiency trade-offs introduced by each quantization configuration.

Model footprint reduction relative to FP16 was calculated as:

$$\text{Footprint Reduction} = \left(1 - \frac{M_{\text{quantized}}}{M_{\text{FP16}}}\right) \times 100 \quad (4)$$

where $M_{\text{quantized}}$ denotes the measured model footprint for INT8 or 4 bit NF4 and M_{FP16} denotes the footprint of the corresponding FP16 model.

This measure quantifies the practical memory savings obtained from compression.

Inference latency measures the elapsed time required to process a standardized input and generate its output. We retain both mean and median latency to account for runtime variation. Throughput is measured as both examples per second and generated tokens per second. These measures are analyzed alongside model footprint because reduced precision may lower memory requirements without

necessarily improving inference speed on a given GPU architecture.

3.7 Performance–Efficiency Trade-off Analysis

The final analysis combines task performance and computational efficiency to examine the trade off introduced by quantization.

For each model configuration, the FP16 baseline was compared with INT8 and 4-bit NF4 using Macro F1, performance retention, absolute F1 change, model footprint, footprint reduction, latency, and generated token throughput.

Performance and memory values were also used to construct trade off analyses that identify configurations providing strong predictive performance while requiring substantially less model memory.

This enables the study to examine compression as a multidimensional problem rather than treating quantization only as a reduction in numerical precision.

4 Results

4.1 Overall Performance and Effect of Quantization

Table 4 and Figure 1 summarize the controlled Qwen2.5 scaling experiment. Under FP16, Macro-F1 increases monotonically with model capacity on all three tasks. Scaling from 0.5B to 3B raises Macro-F1 from 0.085 to 0.435 on stance classification, from 0.165 to 0.728 on natural language inference (NLI), and from 0.132 to 0.664 on fake-news classification. The 1.5B model occupies an intermediate position on stance (0.293) and NLI (0.473), but remains weak on fake news (0.202). These results establish model capacity as a major determinant of absolute Bangla task performance before compression is considered.

The FP16 outputs of Qwen2.5-0.5B indicate severe class collapse. The model predicts a single class for every NLI example, predicts *Fake* for 850 of the 851 fake-news examples, and predicts *Anti-Uprising* for 977 of the 982 stance examples. Its low scores therefore reflect a weak full-precision baseline rather than damage introduced by quantization. This distinction is important because high performance retention for the 0.5B variants does not imply useful absolute performance.

INT8 generally remains close to the corresponding FP16 baseline. For Qwen2.5-3B, INT8 retains 99.9% of FP16 Macro-F1 on stance, 96.2% on NLI,

Model	Stance			NLI			Fake News		
	FP16	INT8	NF4	FP16	INT8	NF4	FP16	INT8	NF4
Qwen2.5-0.5B	0.085	0.090	0.084	0.165	0.165	0.171	0.132	0.133	0.133
Qwen2.5-1.5B	0.293	0.296	0.269	0.473	0.477	0.460	0.202	0.197	0.440
Qwen2.5-3B	0.435	0.434	0.397	0.728	0.700	0.563	0.664	0.639	0.592

Table 4: Macro-F1 across the controlled Qwen2.5 scaling study. Cell shading indicates relative change from the corresponding FP16 baseline, with green indicating numerical increases and darker red indicating greater degradation. Bold indicates the highest Macro-F1 within each model–task group. The unusually large NF4 increase for Qwen2.5-1.5B on fake-news classification is analyzed separately through prediction-distribution analysis. NF4 denotes 4-bit NF4 quantization.

and 96.2% on fake news. Detailed retention results across all model scales and tasks are reported in Appendix C. The corresponding absolute Macro-F1 changes are -0.001 , -0.028 , and -0.025 , respectively. Qwen2.5-1.5B also changes little under INT8: its Macro-F1 changes by $+0.003$ on stance, $+0.004$ on NLI, and -0.005 on fake news. Thus, within the tested Qwen2.5 family, INT8 represents a comparatively conservative compression regime.

The effect of 4-bit NF4 is larger and less uniform. For Qwen2.5-3B, performance retention falls to 91.3% on stance, 77.3% on NLI, and 89.2% on fake news, corresponding to absolute Macro-F1 decreases of 0.038, 0.165, and 0.072, respectively. For Qwen2.5-1.5B, NF4 lowers stance and NLI Macro-F1 to 0.269 and 0.460, but raises fake-news Macro-F1 from 0.202 to 0.440.

Inspection of the prediction distributions (Appendix E) shows that the Qwen2.5-1.5B fake-news result is associated with a substantial change in label bias. Under FP16, the model predicts *Fake* for 789 of the 851 examples. Under NF4, the prediction distribution changes to 475 *Authentic* predictions and 376 *Fake* predictions. The resulting increase in Macro-F1 should therefore be interpreted as a task-specific behavioral shift that happened to produce a more balanced class-level score, rather than evidence that aggressive quantization generally improves model capability.

4.2 Task and Model-Scale Sensitivity

Compression sensitivity varies across both tasks and model scales. For Qwen2.5-3B, the strongest degradation under 4-bit NF4 occurs on NLI, where Macro-F1 decreases from 0.728 to 0.563, corresponding to 22.7% relative degradation. The corresponding reductions are 8.7% for stance and 10.8% for fake news. INT8 produces substantially smaller reductions of 0.1%, 3.8%, and 3.8%, respectively, indicating that task-level sensitivity becomes more

pronounced under aggressive quantization.

Compression robustness also cannot be inferred from retention alone. Although Qwen2.5-0.5B retains approximately 98–103% of its FP16 performance under NF4, these values arise from severely collapsed baselines. In contrast, the losses observed for Qwen2.5-3B occur from substantially stronger baseline performance and therefore represent more meaningful compression-induced degradation. Together with the Qwen2.5-1.5B fake-news prediction shift, these results show that compression sensitivity should be interpreted jointly through absolute performance, retention, and baseline model behavior.

4.3 Cross-Family Validation

Table 5 compares Qwen2.5-3B and Falcon3-3B as a cross-family validation at a matched nominal model scale (See Appendix D.1 for details). This comparison is separate from the controlled Qwen2.5 scaling sequence and does not treat Falcon3 as a fourth point in the Qwen2.5 capacity curve.

Qwen2.5-3B achieves higher absolute Macro-F1 than Falcon3-3B across all tasks and precision settings. Under FP16, the difference is particularly pronounced on stance (0.435 vs. 0.173), while Qwen2.5 also leads on NLI (0.728 vs. 0.491) and fake news (0.664 vs. 0.326).

Falcon3’s FP16 stance predictions are strongly skewed toward *Anti-Uprising*, which is predicted for 832 of the 982 examples. The corresponding per-class F1 scores are 0.012 for *Pro-Uprising*, 0.248 for *Neutral*, and 0.259 for *Anti-Uprising*. Thus, the low overall stance Macro-F1 is associated with a strongly imbalanced prediction distribution rather than a uniform reduction across the three classes.

Both model families preserve most of their FP16 performance under INT8. Falcon3-3B retains

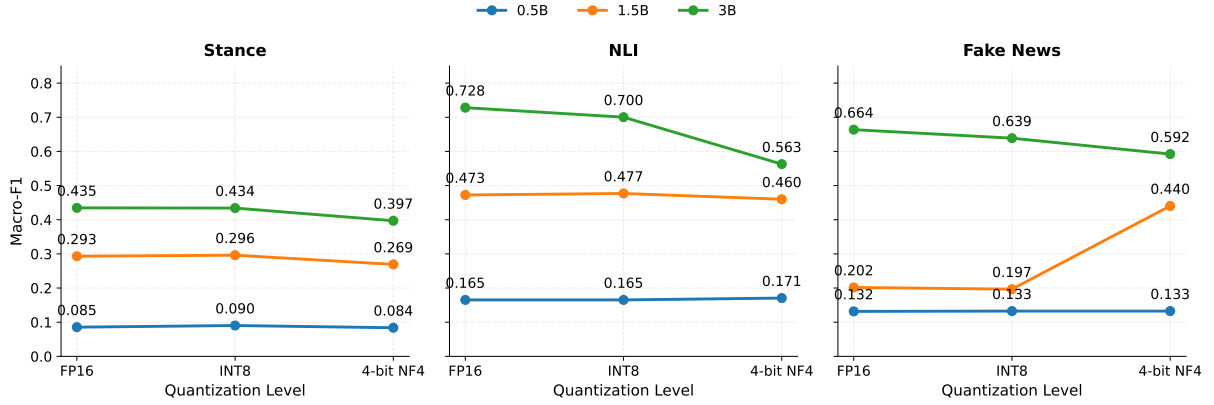


Figure 1: Macro-F1 under FP16, INT8, and 4-bit NF4 across the controlled Qwen2.5 scaling study. Increasing model capacity raises absolute performance on all three tasks, whereas the effect of quantization depends on both model size and task.

93.5% on stance, 98.8% on NLI, and 98.8% on fake news. The corresponding Qwen2.5-3B retention values are 99.9%, 96.2%, and 96.2%. These results suggest that the relative stability of INT8 generalizes across the two evaluated model families, although its effect is not identical for every task.

Under 4-bit NF4, the two families exhibit less consistent task-level behavior. Qwen2.5-3B decreases on all three tasks, whereas Falcon3-3B decreases on NLI but increases numerically on stance from 0.173 to 0.229 and on fake news from 0.326 to 0.482. The Falcon3 stance increase is accompanied by a redistribution of predicted classes and should therefore be interpreted cautiously. On NLI, both models show substantial degradation: Qwen2.5-3B retains 77.3% of its FP16 score, while Falcon3-3B retains 78.8%. Falcon3 fake-news Macro-F1 increases from 0.326 to 0.482, corresponding to 147.7% retention. Overall, INT8 stability generalizes more consistently across the two model families than the task-level behavior of 4-bit NF4.

4.4 Memory and Inference Efficiency

Table 6 reports the standardized efficiency benchmark on the NVIDIA Tesla T4, with complete measurements provided in Appendix F. Model footprint is used as the primary memory measure for comparing precision settings.

For Qwen2.5-3B, INT8 reduces the model footprint from 5.75 to 3.16 GB, corresponding to a 45.0% reduction. The 4-bit NF4 configuration reduces the footprint further to 1.87 GB, corresponding to a 67.4% reduction. Falcon3-3B follows the same ordering: INT8 reduces its footprint from

6.01 to 3.76 GB, a reduction of 37.5%, while NF4 reduces it to 2.63 GB, a reduction of 56.3%. Thus, both quantization settings provide substantial memory savings across model families, with NF4 consistently producing the largest reduction.

The footprint reductions do not translate into faster generation on the Tesla T4. Qwen2.5-3B throughput decreases from 11.91 generated tokens/s under FP16 to 4.18 tokens/s under INT8 and 8.92 tokens/s under NF4. Mean latency simultaneously increases from 0.672 s to 1.914 s under INT8 and 0.897 s under NF4. INT8 therefore reduces model footprint by 45.0% but also reduces generated-token throughput by 64.9%.

Falcon3-3B exhibits the same general memory-speed mismatch. Its throughput decreases from 18.41 tokens/s under FP16 to 7.19 tokens/s under INT8 and 13.73 tokens/s under NF4. Mean latency increases from 0.435 s under FP16 to 1.113 s under INT8 and 0.583 s under NF4. INT8 is the slowest precision setting for both 3B models despite having an intermediate memory footprint. Under the tested hardware and software configuration, quantization therefore functions primarily as a memory optimization rather than a speed optimization.

4.5 Performance–Efficiency Trade-off

Figure 2 summarizes the trade-off between the unweighted mean Macro-F1 across the three tasks and model footprint. Among the Qwen2.5 configurations, 3B INT8 provides a favorable memory–performance trade-off, reducing model footprint from 5.75 to 3.16 GB (45.0%) while retaining 97.4% of the mean FP16 Macro-F1. NF4 further reduces the footprint to 1.87 GB but lowers mean

Model	Stance			NLI			Fake News		
	FP16	INT8	NF4	FP16	INT8	NF4	FP16	INT8	NF4
Qwen2.5-3B	0.435	0.434	0.397	0.728	0.700	0.563	0.664	0.639	0.592
Falcon3-3B	0.173	0.162	0.229	0.491	0.485	0.387	0.326	0.322	0.482

Table 5: Cross-family Macro-F1 comparison between Qwen2.5-3B and Falcon3-3B under FP16, INT8, and 4-bit NF4. Bold indicates the better-performing model within each task–precision comparison.

Model	Precision	Footprint (GB)↓	Red. (%)↑	Latency (s)↓	Tok./s ↑
Qwen2.5-3B	FP16	5.75	0.0	0.672	11.91
	INT8	3.16	45.0	1.914	4.18
	4-bit NF4	1.87	67.4	0.897	8.92
Falcon3-3B	FP16	6.01	0.0	0.435	18.41
	INT8	3.76	37.5	1.113	7.19
	4-bit NF4	2.63	56.3	0.583	13.73

Table 6: Efficiency of the matched 3B models on an NVIDIA Tesla T4. Memory reduction is calculated relative to the FP16 version of each model. Green shading indicates memory savings, while red shading indicates latency or throughput degradation relative to FP16; darker shading represents a larger change.

Macro-F1 from 0.609 to 0.517. In contrast, the smaller Qwen2.5 models require less memory but provide substantially lower absolute task performance, illustrating why retention alone is insufficient for selecting a deployment configuration.

A similar trade-off appears for Falcon3-3B, although at lower absolute performance. Across both families, no precision setting simultaneously optimizes predictive performance, memory footprint, and inference speed. For Qwen 2.5, FP16 provides the strongest performance and throughput on the tested T4, INT8 offers substantial memory reduction with limited performance loss, and NF4 provides the greatest memory savings at the cost of larger and more task-dependent degradation.

5 Conclusion

We presented a controlled study of post-training quantization for small multilingual LLMs on Bangla language understanding, examining the interaction between model capacity, numerical precision, task sensitivity, and computational efficiency. Across Qwen2.5 models from 0.5B to 3B parameters, our results show that baseline model capacity is critical for interpreting compression robustness: high retention can be misleading when the full-precision model already exhibits weak performance or class collapse. INT8 generally preserves FP16

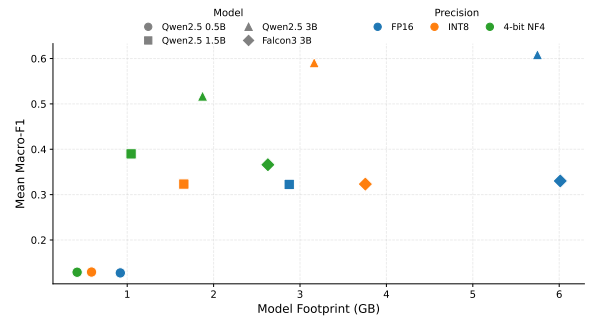


Figure 2: Unweighted mean Macro-F1 across the three evaluated Bangla tasks versus the corresponding model footprint. Each point represents one model–precision configuration; marker shape denotes the model and color denotes the precision setting.

performance with limited degradation, whereas 4-bit NF4 provides greater memory savings but introduces larger and more task-dependent performance changes. The matched Falcon3-3B experiments further demonstrate that compression behavior can vary across model families.

Our efficiency analysis shows that memory reduction does not necessarily translate into faster inference. On an NVIDIA Tesla T4, both INT8 and NF4 substantially reduce model footprint, but neither improves inference speed over FP16. Overall, our findings suggest that practical quantization for resource-constrained Bangla language understanding requires balancing model capability, task sensitivity, memory savings, and hardware-specific inference behavior rather than selecting the lowest numerical precision by default.

6 Limitations

Our study has several limitations. We focus the controlled scaling analysis on three Qwen2.5 model sizes, with Falcon3-3B used as a matched cross-family comparison; additional model families and scales could further assess the generalizability of the observed quantization patterns. We evaluate FP16, INT8, and 4-bit NF4 in prompt-based classification settings, while other quantization methods, numerical formats, open-ended gener-

ation tasks, and fine-tuned models may exhibit different performance–efficiency trade-offs. Our efficiency measurements are also hardware- and implementation-dependent, as the benchmarks are conducted on an NVIDIA Tesla T4 under a fixed inference configuration; latency and throughput may differ across hardware, quantization kernels, and software environments. Finally, we focus on Bangla, and therefore do not assume that the observed behavior generalizes to other low-resource languages. Extending the analysis to additional low- and high-resource languages could help determine whether compression sensitivity varies systematically across languages.

7 Ethical Considerations

The Bangla July Revolution Stance (BJUS) dataset contains naturally occurring social-media comments related to a politically sensitive event. Such content may contain noise, misinformation, or social and political biases, and its use requires careful consideration of the underlying social context. The dataset is used solely for research on aggregate language-understanding performance, and the reported results should not be interpreted as assessments of individual users or their political viewpoints. No user identities, usernames, account identifiers, or other personally identifying information are disclosed in this study. Representative comments are presented without information linking them to their original users, thereby minimizing privacy risks.

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Appendix

A Additional Dataset Details

A.1 BJUS Dataset Details

Dataset provenance. The Bangla July Uprising Stance (BJUS) dataset used in this study is an extended version of the Student-People Mass Uprising Public Sentiments Dataset (Hossen et al., 2025). The original dataset contains 4,200 Bangla social-media comments collected from Facebook, YouTube, and Twitter (X) during and after the July Revolution. The comments were manually collected between September and October, 2024 by searching posts and news related to the event. The original dataset was designed for sentiment analysis and contains an equal distribution of 1,400 positive, 1,400 negative, and 1,400 neutral comments.

Platform	Stance	Examples
YouTube	Pro-Uprising	Comment: একটি যৌক্তিক, শান্তিপূর্ণ, অহিংস আন্দোলন করছিল তখন কেনো আলোচনায় বসলেন না? এতো হত্যাযজ্ঞ চালালেন কেনো? (Why did you not engage in meeting when a logical, peaceful, non-violent movement was taking place? Why did you carry out so much killing?)
Facebook	Pro-Uprising	Comment: কোন সংলাপ হবে না। খুনি হাসিনার পদত্যাগ চাই এবং কঠোর শাস্তি চাই। (There will be no conversation. We demand the resignation of “killer Hasina” and severe punishment.)
YouTube	Anti-Uprising	Comment: সব কয়টা রাজনৈতিক করে,এরা কেউ সাধারণ শিক্ষার্থী নয় (They are all involved in politics; none of them are ordinary students.)
Facebook	Neutral	Comment: অপরাধের কোন ধর্ম দলীয় পরিচয় নেই। অপরাধী রাষ্ট্রীয় আইনের কাঠামোয় বিচার হবে এর জন্য দল দায়ভার নেবে না। (Crime has no religious or political identity. The offender will be tried under state law; the party will not bear responsibility for it.)
YouTube	Anti-Uprising	Comment: কোটি টাকা না পাইলে কি আর এখনো মাঠে আছে কোটা আন্দোলন কারীরা? (Would the quota protesters still be on the streets if they had not received crores of money?)

Table 7: Representative examples from the BJUS dataset. Each Bangla comment is accompanied by an English translation in parentheses.

Original annotation scheme. In the original dataset, comments expressing support for the protest or containing positive remarks were labeled as *Positive*. Comments containing offensive expressions or threats directed toward the protesting students were categorized as *Negative*, while comments that neither expressed support nor contained such language were labeled as *Neutral*. The resulting annotations were organized in a structured CSV format.

Extension and stance annotation. For the present study, the original resource was extended with additional Bangla comments and manually assigned stance annotations. The task was formulated as stance classification rather than sentiment classification because sentiment polarity does not necessarily correspond to a comment’s position toward the uprising. Each comment was assigned one of three stance labels based on its expressed position: *Pro-Uprising* for support, *Anti-Uprising* for opposition, and *Neutral* when no clear position toward either side was expressed. The stance annotations were assigned separately from the original sentiment labels.

Table 7 presents representative BJUS comments together with their English translations and stance annotations. Only the stance annotations are used as prediction targets in the experiments reported in this study.

A.2 Data Preprocessing

We apply task-specific preprocessing to construct standardized evaluation datasets for BJUS, XNLI-BN, and BanFakeNews. We preserve the origi-

nal Bangla text rather than applying conventional NLP preprocessing such as stemming, stop-word removal, or punctuation removal, since the evaluated LLMs operate directly on natural-language inputs. The processed datasets are subsequently divided into fixed train, development, and test partitions using a random seed of 42. Only the fixed test partitions are used for the zero-shot evaluations reported in this study.

Table 1 in the main paper summarizes the resulting fixed data partitions. These partitions are established before evaluation and remain unchanged across model sizes, model families, and quantization settings.

BJUS. For the stance task, we use Bangla social-media comments from the extended BJUS dataset and formulate the target as three-way stance classification. The stance annotations are standardized into *Pro-Uprising*, *Neutral*, and *Anti-Uprising*. The resulting dataset contains 9,817 examples. We construct stratified train, development, and test partitions containing 7,853, 982, and 982 examples, respectively. The corresponding class distributions are 2,733/2,699/2,421 for Pro-Uprising, Neutral, and Anti-Uprising in the training set; 342/337/303 in the development set; and 341/338/303 in the test set. The social-media comment itself serves as the model input, while the stance annotation serves as the gold output label.

XNLI-BN. For natural language inference, each example consists of a Bangla premise–hypothesis pair and one of three labels: *Entailment*, *Neutral*, or *Contradiction*.

The premise and hypothesis are retained as separate fields during preprocessing and are combined only when constructing the model prompt.

Because multiple NLI instances can originate from the same underlying example group, we preserve the associated group identifiers and perform group-aware splitting to prevent related examples from appearing across different partitions. Specifically, the data are partitioned using group-level splitting with a random seed of 42, with no group overlap between train, development, and test sets. The final dataset contains 12,528 examples, divided into 10,024 training, 1,251 development, and 1,253 test examples. The test partition contains 420 Neutral, 420 Contradiction, and 413 Entailment examples.

BanFakeNews. For fake-news detection, the article headline and content are retained as separate fields during preprocessing and are combined during prompt construction. The binary authenticity annotation is mapped to the output labels *Authentic* and *Fake*. Only the manually annotated portion of the larger BanFakeNews corpus is used. During dataset preparation, article identifiers are checked across source files to prevent incorrect global duplicate removal. After preprocessing, 8,501 examples remain, comprising 7,202 authentic and 1,299 fake articles. Stratified splitting with a random seed of 42 produces 6,800 training, 850 development, and 851 test examples, with 721 authentic and 130 fake articles in the test set.

Long-input handling is performed after prompt construction and is described in Appendix B, since truncation is part of the inference protocol rather than dataset preprocessing.

B Prompting and Inference Protocol

We use a unified constrained-generation framework for all three tasks. Each input is converted into a task-specific instruction prompt that specifies the permissible output labels and requests a single label as the response. For a given task, the prompt template is kept identical across model families, model sizes, and quantization configurations. This ensures that comparisons are not affected by differences in prompt formulation.

B.1 Task-Specific Prompt Construction

The three tasks differ in their task instruction, input fields, and permissible output labels. Stance classification uses the Bangla social-media comment

as input, NLI uses a premise–hypothesis pair, and fake-news detection uses the article headline and content. Table 8 summarizes the corresponding input fields and output labels.

Task	Input	Allowed Labels
Stance	Comment	Pro-Uprising, Neutral, Anti-Uprising
NLI	Premise + Hypothesis	Entailment, Neutral, Contradiction
Fake News	Headline + Content	Authentic, Fake

Table 8: Input fields and permissible output labels for each evaluation task.

B.2 Prompt Templates

The prompt templates used for evaluation are provided below. The placeholders are replaced with the corresponding fields of each test example before applying the model-specific chat template.

Stance. The stance prompt specifies the position to be classified and provides definitions for each of the three permissible stance labels.

Determine the stance of the following Bangla social media comment toward the July Revolution in Bangladesh.

Use these labels:
 Pro-Uprising: the comment supports the uprising, protesters, or protesting students, or expresses opposition to actions against them.
 Anti-Uprising: the comment opposes, criticizes, or attacks the uprising, protesters, or protesting students, or supports actions against them.
 Neutral: the comment does not express a clear position either for or against the uprising.

Choose exactly one label: Pro-Uprising, Anti-Uprising, Neutral.
 Return only the label.

Text: {text}

Natural Language Inference. For NLI, the premise and hypothesis are provided separately and the model is restricted to one of three permissible labels.

Determine the relationship between the following Bangla premise and hypothesis.
 Choose exactly one label: Entailment, Neutral, Contradiction.
 Return only the label.

Model	Task	Precision	Acc.	Macro-P	Macro-R	Macro-F1	Parse (%)
Qwen2.5-0.5B	Stance	FP16	.144	.214	.330	.085	100.0
	Stance	INT8	.147	.436	.332	.090	100.0
	Stance	NF4	.144	.048	.333	.084	100.0
	NLI	FP16	.330	.110	.333	.165	100.0
	NLI	INT8	.330	.110	.333	.165	100.0
	NLI	NF4	.332	.360	.336	.171	100.0
	Fake News	FP16	.152	.076	.496	.132	100.0
	Fake News	INT8	.153	.076	.500	.133	100.0
	Fake News	NF4	.153	.076	.500	.133	100.0
Qwen2.5-1.5B	Stance	FP16	.379	.355	.374	.293	100.0
	Stance	INT8	.395	.344	.359	.296	100.0
	Stance	NF4	.355	.236	.390	.269	100.0
	NLI	FP16	.581	.621	.583	.473	100.0
	NLI	INT8	.585	.651	.587	.477	100.0
	NLI	NF4	.564	.570	.566	.460	100.0
	Fake News	FP16	.207	.513	.507	.202	100.0
	Fake News	INT8	.203	.531	.514	.197	100.0
	Fake News	NF4	.532	.492	.484	.440	100.0
Qwen2.5-3B	Stance	FP16	.449	.432	.466	.435	100.0
	Stance	INT8	.451	.431	.466	.434	100.0
	Stance	NF4	.432	.425	.413	.397	100.0
	NLI	FP16	.725	.768	.725	.728	100.0
	NLI	INT8	.706	.761	.706	.700	100.0
	NLI	NF4	.598	.734	.598	.563	100.0
	Fake News	FP16	.765	.649	.742	.664	100.0
	Fake News	INT8	.730	.635	.736	.639	100.0
	Fake News	NF4	.658	.622	.732	.592	100.0

Table 9: Additional evaluation metrics for the controlled Qwen2.5 experiments. Macro-P, Macro-R, and Macro-F1 denote macro-averaged precision, recall, and F1, respectively. Parse denotes the percentage of generated responses successfully mapped to a valid task label.

Premise: {Premise}
Hypothesis: {Hypothesis}

Fake-News Detection. For fake-news detection, the processed article text is provided as a single input field and the response is restricted to one of two labels.

Classify the following Bangla news article.
Choose exactly one label: Authentic, Fake.
Return only the label.

Article: {text}

B.3 Decoding and Inference

Inference is deterministic for all experimental configurations. Sampling is disabled using `do_sample=False`, and generation is limited to `max_new_tokens=8`. Each model receives the same test examples in the same processed form across FP16, INT8, and 4-bit NF4 configurations. The corresponding model tokenizer and chat template are used to construct the final model input.

The maximum input length is fixed at 4096 tokens. Token length is measured after constructing the complete prompt so that the task instruction and input fields are jointly considered when enforcing the context limit. This constraint primarily affects the longer fake-news inputs.

B.4 Input Truncation

Inputs exceeding the maximum context length are truncated using a controlled procedure rather than truncating the complete formatted prompt indiscriminately. The task instruction, permissible output labels, and prompt structure are preserved, while tokens are removed from the variable-length input content until the complete prompt fits within the specified token budget. The same truncation procedure is applied across model and quantization configurations.

For each evaluated example, we record the original input-token count, the number of tokens retained after truncation, and whether truncation was required. This allows truncated examples to be identified during subsequent analysis.

B.5 Output Parsing

Generated responses are processed using deterministic task-specific parsers. For each task, the parser attempts to map the generated response to one of its predefined permissible labels. Valid responses are converted to the corresponding canonical label before metric computation.

Outputs that cannot be mapped to a permissible label are recorded as parsing failures rather than being assigned to another class. For every test example, we retain the gold label, predicted label,

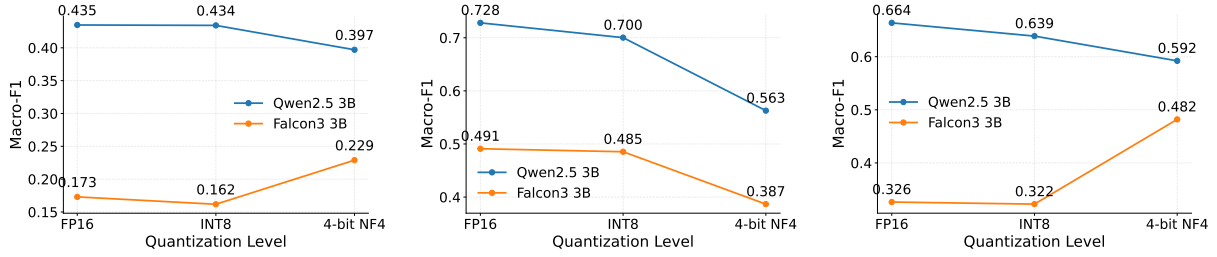


Figure 3: Task-specific cross-family comparison between Qwen2.5-3B and Falcon3-3B under FP16, INT8, and 4-bit NF4 quantization for stance (left), NLI (center), and fake-news classification (right). Compression behavior varies across both tasks and model families.

raw generated response, and parsing-success status. This procedure is kept identical across model sizes, model families, and quantization configurations.

B.6 Reproducibility and Experiment Management

All experiments use a common configuration-driven pipeline, with model, dataset, inference, and experiment settings separated from the evaluation logic. The same evaluation code is used across model sizes and precision levels.

Raw predictions are stored for each model–task–quantization configuration and used to compute aggregate metrics and generate summary tables, retention and sensitivity analyses, efficiency summaries, and figures. The implementation separates configuration, data preparation, model loading, evaluation, and analysis into modular components. The software environment is reported in Table 10.

Component	Version / Platform
Python	3.12.13
PyTorch	2.10.0+cu128
Transformers	5.0.0
bitsandbytes	0.50.2
Accelerate	1.13.0
CUDA	12.8
Environment	Kaggle

Table 10: Experimental Software environment.

C Detailed Performance Retention

Table 11 reports Macro-F1 retention for the Qwen2.5 models across all tasks and quantization settings. Retention is calculated relative to the FP16 performance of the same model and task. Values above 100% indicate a numerical increase over the corresponding FP16 baseline and should therefore be interpreted together with absolute task performance.

Figure 4 provides a complementary visualization of Table 11. INT8 remains close to the corre-

Model	Task	INT8 (%)	NF4 (%)
Qwen2.5-0.5B	Stance	105.6	97.9
	NLI	100.0	103.2
	Fake News	100.7	100.7
Qwen2.5-1.5B	Stance	101.0	91.8
	NLI	100.9	97.4
	Fake News	97.5	218.3
Qwen2.5-3B	Stance	99.9	91.3
	NLI	96.2	77.3
	Fake News	96.2	89.2

Table 11: Qwen2.5 Macro-F1 retention relative to the FP16 version of the same model. Retention above 100% denotes a numerical increase relative to the corresponding FP16 score and does not necessarily indicate improved model capability. NF4 denotes 4-bit NF4 quantization.

sponding FP16 baseline across most configurations, whereas NF4 shows greater variation across model sizes and tasks. Qwen2.5-3B exhibits its largest NF4 degradation on NLI, while the unusually high retention of Qwen2.5-1.5B on fake-news classification is examined further in Appendix E.

Figure 5 isolates Qwen2.5-3B to examine task-specific sensitivity at the strongest model scale in the controlled study. INT8 produces limited changes across the three tasks, whereas NF4 causes larger degradation, particularly for NLI. This demonstrates that compression robustness remains task-dependent even when model architecture and parameter scale are fixed.

D Additional Evaluation Results

Macro-F1 is used as the primary evaluation metric in the main paper. For completeness, Table 9 reports Accuracy, Macro Precision, Macro Recall, Macro-F1, and parsing success for all Qwen2.5 configurations. Parsing success is 100% across these configurations, indicating that the observed performance differences are not attributable to unparsable generations.

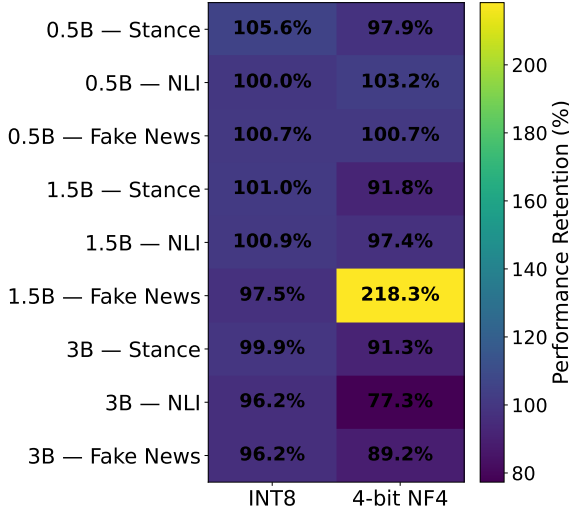


Figure 4: Macro-F1 retention relative to the corresponding FP16 baseline across Qwen2.5 model sizes, tasks, and quantization levels. Values near 100% indicate limited change relative to FP16, while lower values indicate greater degradation. Values above 100% represent numerical increases over the corresponding FP16 score.

D.1 Cross-Family Analysis

Figure 3 provides a task-level view of the matched 3B cross-family comparison. The effect of quantization varies across both model families and tasks, and Falcon3 does not exhibit a uniform monotonic decline as precision is reduced. This reinforces that matched parameter scale does not imply identical sensitivity to quantization.

E Prediction Distribution Analysis

Aggregate metrics can obscure highly concentrated prediction behavior, particularly when the full-precision baseline is already weak. Table 12 therefore reports prediction distributions for Qwen2.5-0.5B, which exhibits severe class concentration, and Qwen2.5-1.5B on fake-news classification, where NF4 produces an unusually large increase in Macro-F1.

E.1 Qwen2.5-0.5B Class Collapse

The Qwen2.5-0.5B results demonstrate why high relative retention does not necessarily imply useful task capability. Under stance NF4, all 982 test examples are assigned to a single class. For NLI, FP16 and INT8 assign all 1,253 examples to the same class, while NF4 remains similarly concentrated. Fake-news predictions are also almost entirely concentrated in one class. Consequently, the near-100% retention values in Table 11 largely pre-

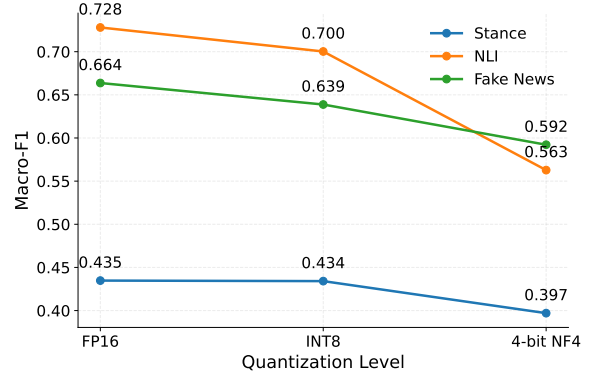


Figure 5: Task-specific compression sensitivity of Qwen2.5-3B under FP16, INT8, and 4-bit NF4. NLI exhibits the largest degradation under NF4, while stance and fake-news classification are comparatively less affected.

Model	Task	Precision	Predicted Labels		
			Class 1	Class 2	Class 3
Qwen2.5-0.5B	Stance	FP16	4	1	977
	Stance	INT8	6	2	974
	Stance	NF4	0	0	982
Qwen2.5-0.5B	NLI	FP16	1253	0	0
	NLI	INT8	1253	0	0
	NLI	NF4	1247	2	4
Qwen2.5-0.5B	Fake News	FP16	1	850	–
	Fake News	INT8	0	851	–
	Fake News	NF4	0	851	–
Qwen2.5-1.5B	Fake News	FP16	62	789	–
	Fake News	INT8	53	798	–
	Fake News	NF4	475	376	–

Table 12: Predicted-label distributions for selected diagnostic configurations. For stance, Classes 1–3 correspond to Pro-Uprising, Neutral, and Anti-Uprising; for NLI, they correspond to Entailment, Neutral, and Contradiction; for fake news, Classes 1–2 correspond to Authentic and Fake.

serve an already weak baseline rather than demonstrating robust compression.

E.2 Qwen2.5-1.5B Fake-News NF4 Shift

Qwen2.5-1.5B exhibits a markedly different pattern on fake-news classification. FP16 predicts the two classes 62 and 789 times, respectively, and INT8 produces a similar distribution of 53 and 798. Under NF4, however, the distribution shifts substantially to 475 and 376 predictions. This redistribution coincides with the increase in Macro-F1 from 0.202 under FP16 to 0.440 under NF4 and the resulting 218.3% retention reported in Table 11. We therefore interpret this result as a quantization-induced change in class preference rather than evidence that aggressive quantization generally improves model capability.

Model	Prec.	Load (s)	Foot. (GB)	Peak (GB)	Mean Lat. (s)	Median Lat. (s)	Ex./s	Tok./s	Red. (%)
Qwen2.5-0.5B	FP16	33.195	0.920	0.502	0.407	0.405	2.456	19.650	0.0
	INT8	4.716	0.587	0.319	1.264	1.259	0.791	6.327	36.2
	NF4	1.901	0.420	0.009	0.402	0.400	2.485	19.878	54.3
Qwen2.5-1.5B	FP16	28.293	2.875	1.454	0.510	0.507	1.960	15.679	0.0
	INT8	3.880	1.655	0.813	1.472	1.468	0.679	5.434	42.4
	NF4	3.488	1.045	0.545	0.620	0.619	1.613	12.903	63.7
Qwen2.5-3B	FP16	87.072	5.748	2.954	0.672	0.668	1.489	11.911	0.0
	INT8	7.524	3.164	1.524	1.914	1.914	0.522	4.179	45.0
	NF4	7.479	1.872	0.949	0.897	0.894	1.115	8.918	67.4
Falcon3-3B	FP16	184.407	6.012	2.483	0.435	0.434	2.301	18.406	0.0
	INT8	9.838	3.756	1.361	1.113	1.111	0.899	7.190	37.5
	NF4	7.051	2.628	0.897	0.583	0.583	1.716	13.730	56.3

Table 13: Complete efficiency measurements on an NVIDIA Tesla T4. Load denotes model loading time, Foot. denotes model footprint, Peak denotes peak inference memory, Mean Lat. and Median Lat. denote inference latency, Ex./s denotes examples per second, Tok./s denotes generated tokens per second, and Red. denotes model-footprint reduction relative to FP16 for the same model. Green shading indicates memory or throughput improvements relative to FP16, whereas red shading indicates inference-efficiency degradation; darker shading represents a larger change.

Setting	Value
Decoding	Greedy / deterministic
Sampling	Disabled
Max. new tokens	8
Max. input tokens	4,096
Batch size (benchmark)	1
Benchmark input length	~512 tokens
Warm-up runs	3
Measured runs	20
Hardware	NVIDIA Tesla T4

Table 14: Inference and efficiency-benchmark configuration used for the standardized measurements.

F Additional Efficiency Results

Efficiency measurements are conducted on an NVIDIA Tesla T4 using a standardized inference configuration. Table 14 summarizes the benchmark protocol. All measurements use batch size 1 and approximately 512 input tokens, with three warm-up runs followed by 20 measured runs and CUDA synchronization during timing.

Table 13 reports the complete efficiency measurements across model and precision configurations. In addition to model footprint and mean latency reported in the main paper, the table includes model loading time, peak inference memory, median latency, examples per second, and generated-token throughput.

Figure 6 summarizes the memory–throughput trade-off across Qwen2.5 model scales. INT8 and NF4 substantially reduce model footprint, with the absolute savings increasing with model size. For

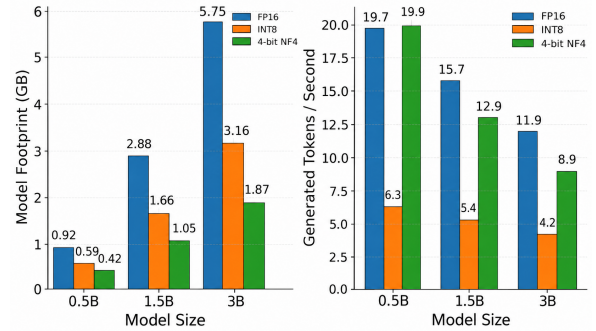


Figure 6: Memory and inference-throughput effects of quantization across Qwen2.5 model scales. INT8 and 4-bit NF4 substantially reduce model footprint relative to FP16, while these reductions do not necessarily translate into higher generated-token throughput on the NVIDIA Tesla T4.

Qwen2.5-3B, the footprint decreases from 5.75 GB under FP16 to 3.16 GB under INT8 and 1.87 GB under NF4, corresponding to reductions of approximately 45.0% and 67.4%, respectively. However, these memory reductions do not translate directly into higher inference throughput on the evaluated hardware. For Qwen2.5-3B, generated-token throughput decreases from 11.91 tokens/s under FP16 to 4.18 tokens/s under INT8, while NF4 reaches 8.92 tokens/s. Overall, these results demonstrate that lower-precision quantization provides substantial memory savings without necessarily improving inference speed.